Class 7

Chemistry

(Solutions)

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1. Acids, Bases and Salts

Solutions

LEVEL-1

1. Correct option: (A)
The acid present in stomach for digestion is hydrochloric acid, which is also known as digestive acid.
2. Correct option: (B)
Sodium Hydrocarbonate used for baking bread. It is also known as baking soda.
3. Correct option: (B)
The formula for common salt used daily is sodium chloride (NaCl).
4. Correct option: (B)
The indicators tell about the acidic nature of a substance by change the colour when they mix with a substance.
5. Correct option: (B)
An indicator gives different colours in acid and base.
6. Correct option: (C)
The colour of the litmus solution when it is neutral is mauve(pale purple).
7. Correct option: (A)
The colour of the litmus solution when an acid is added is red.
8. Correct option: (B)
The colour of the litmus solution when a base is added is blue.
9. Correct option: (B)
Lichen is the name of the plant from which litmus is extracted.
10. Correct option: (A)
If a drop of the given solution turns blue litmus paper to red, then the given solution will be acid in nature.
11. Correct option: (B)
If a drop of the given solution turns red litmus to blue, then the given solution will be base in nature.
12. Correct option: (A)
In Hindi china indicator is called as Gudhal.
13. Correct option: (B)
Light pink is the colour of the China rose solution.
14. Correct option: (A)
Acid turns China rose solution from light pink to deep pink.
15. Correct option: (C)
Bases turns China rose from light pink to green.
16. Correct option: (C)
Turmeric turns red in basic solution.
17. Correct option: (D)
Turmeric remains yellow in acidic solution.
18. Correct option: (A)
Phenolphthalein is a man made indicator.
19. Correct option: (C)
Phenolphthalein indicator is colourless in acid solution.
20. Correct option: (D)
Phenolphthalein indicator is pink in basic solution.

21. Correct option: (D)
The acid present in lemon, which gives it a sour taste, is called citric acid.
22. Correct option: (A)
The acid present in plant material and animals are natural acids called organic acids.
23. Correct option: (C)
An acid reacts with a base to form a salt and water.
24. Correct option: (A)
Acids have sour taste.
25. Correct option: (B)
Bases have bitter taste.
26. Correct option: (C)
Acids are divided into organic and mineral acids.
27. Correct option: (A)
Acetic acid is present in vinegar.
28. Correct option: (D)
Formic acid is present in ant's sting.
29. Correct option: (B)
Citric acid is present in citrus fruits such as lemons and oranges.
30. Correct option: (C)
Lactic acid is present in curd and in sour milk.

LEVEL-II

1. Correct option: (D)
Tartaric acid is present in tamarind, unripe grapes and unripe mangoes.
2. Correct option: (B)
Ascorbic acid and citric acid are present in amla and citrus fruits.
3. Correct option: (C)
Ascorbic acid and citric acid are commonly known as VitaminC.
4. Correct option: (C)
Oxalic acid present in spinach.
5. Correct option: (C)
The taste of bases is bitter.
6. Correct option: (C)
Bases have a slippery touch.
7. Correct option: (B)
Bases are the chemical opposites of acids.
8. Correct option: (B)
All metal hydroxides are bases.
9. Correct option: (A)
The base that is soluble in water is called alkali.
10. Correct option: (C)
Calcium hydroxide found in lime water.
11. Correct option: (D)
Magnesium hydroxide found in milk of magnesia.
12. Correct option: (D)
Ammonium hydroxide found in window cleaner.
13. Correct option: (A)
Sodium carbonate is found in washing soda.

14. Correct option: (A)
Sodium hydrogen carbonate is found in baking soda.
15. Correct option: (D)
All the given compounds are examples of weak bases. The solutions of calcium hydroxide and sodium carbonate are unsafe to drink. But the solution of magnesium hydroxide is safe to drink. it is used for indigestion mixtures like milk of magnesia.
16. Correct option: (A)
Sodium hydroxide is the chemical name of caustic soda.
17. Correct option: (C)
Those substances which are neither acidic nor basic in nature are called neutral substances.
18. Correct option: (C)
The reaction in which an acid reacts with a base to form salt and water is called neutralization.
19. Correct option: (A)
Heat form of energy evolved in a neutralization reaction.
20. Correct option: (A)
NaCl is the chemical formula of the salt formed in this neutralization reaction of sodium hydroxide and hydrochloric acid.
21. Correct option: (C)
Salt and water are the substances formed by the reaction of an acid with a base.
22. Correct option: (C)
Acid and base is the parent compound of a salt.
23. Correct option: (B)
The first part of the salt's name is derived from the name of the base.
24. Correct option: (A)
The second part of the salt's name is derived from the name of the acid.
25. Correct option: (D)
The salts of hydrochloric acid are called chlorides.
26. Correct option: (C)
The salts of nitric acid are called nitrates.
27. Correct option: (A)
Acidic salts turns blue litmus to red. These salts are formed by the reaction of strong acid and weak base. So the total hydrogen are not replaced by the base in neutralization reaction.

2. Classification of Elements

Solutions

LEVEL-I

1. Correct option: (C)

The electro positivity of elements defined as the tendency to lose electrons and form positive ions.

Sodium (Na) is a highly electropositive alkali metal with atomic number 11. It readily loses one electron to form a sodium ion (Na^+). Iron and Nickel are transition metals and are less electropositive than sodium. Chlorine is a non-metal and is highly electronegative, that means it tends to gain electrons, making it the least electropositive element among the options.

2. Correct option: (D)

Electro negativity of an element is defined as the ability to attract and hold onto electrons when chemical bond is formed. Chlorine is a halogen, with atomic number 17 near to Argon (inert gas) and halogens are known for their high electro negativity. They have a strong tendency to attract electrons in chemical bonds, making them highly electronegative. Oxygen is also quite electronegative and comes next after chlorine

Nitrogen and carbon are less electronegative compared to chlorine and oxygen, with carbon generally having lower electro negativity among the two.

3. Correct option: (B)

Antoine Lavoisier, a French chemist, is considered the "Father of Modern Chemistry." his work laid the foundation for the systematic study of elements and their properties. Lavoisier's experiments and studies in the late 18th century helped to establish the concept of chemical elements and the distinction between metals and non-metals based on their properties.

4. Correct option: (D)

Carbon, Nitrogen, and Oxygen can form a triad in the concept proposed by Johann Wolfgang Dobereiner's. They have similar properties, and their atomic masses follow a pattern where the atomic mass of the middle element (Nitrogen) is approximately the average of the atomic masses of the other two elements (Carbon and Oxygen).

5. Correct option: (A)

The law of triads was indeed proposed by Johann Wolfgang Dobereiner, he is a German chemist, in 1829.He noticed that certain groups of three elements had similar chemical properties, and the atomic mass of the middle element was approximately the average of the other two elements in the triad. This observation was the basis of his concept of triads.

6. Correct option: (B)

The systematic relationship between the order of atomic masses and the repetition of properties of elements is called "Periodicity."It refers to the regular and predictable pattern of chemical and physical properties that occur as you move across a row (period) or down a column (group) of the periodic table.

The periodicity is a result of the elements being arranged in order of increasing atomic number, and this arrangement allows elements with similar electron configurations and valence electron arrangements to be placed together, leading to the repetition of properties at regular intervals.

7. Correct option: (A)
John Newlands, an English chemist, proposed the law of octaves in 1864. He observed that when elements were arranged in order of increasing atomic masses, their properties repeated every eighth element, similar to the repetition of musical notes in an octave. According to this law, potassium is the eighth element after sodium in the periodic table, and its properties are similar to those of sodium.

8. Correct option: (A)
Elements are arranged in order of increasing atomic masses and observed that certain properties repeated every eighth element. In this case, fluorine is the first element, and the eighth element after fluorine is chlorine. The properties of chlorine are similar to those of fluorine, following the pattern set by Newlands' law of octaves and they also belongs to the same group halogens.

9. Correct option: (C)
John Newlands' law of octaves, proposed in 1864, he done early attempt to classify the elements based on their properties. However, it was limited in its applicability as it worked well only up to calcium. Beyond calcium, the pattern of repetition of properties every eighth element breaks down, and the law becomes less accurate in describing the properties of elements.

10. Correct option: (D)
John Newlands proposed the law of octaves in 1864, which was the first attempt to classify the elements based on their properties and atomic masses. He noticed that when elements were arranged in order of increasing atomic masses, certain properties repeated every eighth element, similar to the repetition of musical notes in an octave. While Newlands' law of octaves had limitations, it was a crucial step towards the development of the modern periodic table.

11. Correct option: (A)
John Newlands proposed the law of octaves in 1864; he arranged the elements in order of increasing atomic masses and noticed that certain properties repeated every eighth element. Starting from nitrogen (atomic number 7), the eighth element after it would be phosphorus (atomic number 15).

12. Correct option: (C)
Ruthenium (Ru), Rhodium (Rh), and Palladium (Pd) do not form a Dobereiner triad because their atomic masses do not exhibit the required pattern of the atomic mass of the middle element being the average of the other two. Dobereiner's triad concept was more successful in organizing elements in the early days of chemistry but later it was replaced by Mendeleev periodic table and others.

13. Correct option: (B)
John Newlands arranged the elements in increasing order of their atomic numbers. In his law of octaves, he observed that when elements were arranged in order of increasing atomic masses, certain properties repeated every eighth element, similar to the repetition of musical notes in an octave.

14. Correct option: (C)
A set of three elements in which the atomic mass of the middle element is approximately the average of the atomic masses of the other two elements is known as a "Triad." This concept was proposed by Johann Wolfgang Döbereiner, a German chemist, in the early 19th century.

15. Correct option: (A)
The discovery of the inert gases, also known as noble gases, gave a death blow to Newland's law of octaves. Newland's law of octaves was an early attempt to classify the

elements based on their properties and atomic masses. However, the law was limited in its applicability and accuracy.

When the noble gases were discovered in the early 20th centuries (helium, neon, argon, krypton, xenon, and radon), they did not fit into the pattern described by Newland's law of octaves. The noble gases have unique properties, such as being chemically inert and having a full valence electron shell, which makes them distinct from other elements.

16. Correct option: (B)

Rutherford is not associated with the development of the periodic table. The periodic table was first proposed by Dmitri Mendeleev in 1869, and it has since undergone various modifications and refinements by multiple scientists, including Newlands and Lother Mayer. Ernest Rutherford, on the other hand, made significant contributions to the understanding of atomic structure and is known for his experiments that led to the discovery of the atomic nucleus.

17. Correct option: (A)

John Newlands, an English chemist, he observed the periodicity of elements based on their properties. In 1865, he proposed the "Law of Octaves," which stated that when the elements were arranged in order of increasing atomic weights, every eighth element had properties similar to the first, much like the octaves in music.

18. Correct option: (B)

Lother Meyer plotted the graph between atomic volume and atomic mass on X and Y axis respectively. This graph is known as the "Meyer's Curve" or "Meyer's Plot" and is an important contribution to the early understanding of periodic trends in the properties of elements.

In the plot, the x-axis represents the atomic volume, which and is typically expressed in units like cubic centimeters per mole ($\frac{cm^3}{mole}$). The y-axis represents the atomic mass of the elements, usually given in atomic mass units (amu).

19. Correct option: (A)

Lothar Meyer's curve, the peaks are occupied by alkali metals. The curve is also known as the "Meyer's Curve" and it illustrates the relationship between atomic volume (molar volume) and atomic mass of elements.

Alkali metals are found in the leftmost column of the periodic table (Group 1). They include elements such as lithium (Li), sodium (Na), potassium (K), and others. These elements have relatively large atomic volumes and high electropositive nature which is why they appear as peaks on the graph.

The graph helps to show the periodicity of elements, and in Meyer's case, it revealed that elements with similar properties tend to occur at regular intervals when they are arranged by increasing atomic mass.

20. Correct option: (C)

Halogens are found in the 17th group (Group 17) of the periodic table includes elements such as fluorine (F), chlorine (Cl), bromine (Br), iodine (I), and astatine (At). These elements have relatively smaller atomic volumes compared to their atomic masses and they are highly electronegative in nature, which is why they appear in ascending positions on the graph.

21. Correct option: (B)

Alkaline earth metals are found in the IInd group of the periodic table and include elements such as Beryllium (Be), Magnesium (Mg), Calcium (Ca), Strontium (Sr), Barium

(Ba), and Radium (Ra). These elements have relatively larger atomic volumes compared to their atomic masses, which is why they appear in descending positions on the graph.

22. Correct option: (C)

The minimum curve of Lothar Meyer's curve is occupied by the transition metals. The curve, also known as the "Meyer's Curve" represents the relationship between atomic volume (molar volume) and atomic mass of elements.

Transition metals are found in the d-block of the periodic table and include elements such as Iron (Fe), Copper (Cu), Zinc (Zn), Silver (Ag), and Gold (Au), among others. These elements have relatively smaller atomic volumes compared to their atomic masses, which is why they appear at the minimum positions on the graph.

23. Correct option: (A)

Alkali metals are found in Group 1 of the periodic table and include elements such as Lithium (Li), Sodium (Na), Potassium (K), and others. These elements have relatively large atomic volumes compared to their atomic masses, which is why they appear as peaks on the Meyer's curves. The graph helps to demonstrate the periodicity of elements.

24. Correct option: (B)

Dmitri Mendeleev made the most significant contribution to the development of the periodic table. In 1869, Mendeleev published his periodic table, arranging the known chemical elements in a tabular form based on their atomic masses and chemical properties. He left gaps for elements that were yet to be discovered, predicting their properties based on the trends he observed in the table.

Mendeleev's periodic table was groundbreaking because he not only organized the elements according to their atomic masses but also recognized that elements with similar properties occurred at regular intervals. He used the periodicity of the elements to group them into columns (groups) and rows (periods) and laid the foundation for the modern periodic table we use today.

25. Correct option: (B)

Mendeleev organized the elements according to their atomic weights (atomic masses) as the primary sorting principle. He arranged the elements in order of increasing atomic weights and observed that elements with similar properties appeared at regular intervals.

26. Correct option: (C)

Dmitri Mendeleev's periodic table initially included 63 known elements, which he arranged based on their atomic weights and chemical properties. He left gaps in the table for elements that were yet to be discovered, predicting their properties based on the trends he observed in the known elements.

27. Correct option: (D)

Both pair 'B' (Cobalt, Nickel) and pair 'C' (Argon, Potassium) are anomalous pairs in the context of the periodic table.

Pair 'B' (Cobalt, Nickel): Cobalt (Co) and Nickel (Ni) are transition metals and are located close to each other in the d-block of the periodic table. However, their atomic masses are not in strict accordance with the general trend of increasing atomic masses as we move from left to right across the periodic table. Nickel (Ni) has a slightly greater atomic mass than Cobalt (Co), which is anomalous in the typical progression of atomic masses across the periodic table.

Pair 'C' (Argon, Potassium): Argon (Ar) and Potassium (K) are located in Group 18 (noble gases) and Group 1 (alkali metals), respectively. Usually, elements within the same group tend to have a significant difference in their atomic masses. However, Argon (Ar) and Potassium (K) have atomic masses that are relatively close to each other, which is anomalous compared to the general trend observed in the periodic table.

28. Correct option: (B)
Groups.
29. Correct option: (B)
30. Correct option: (D)
According to Mendeleev's periodic table Eka Boron is later named as Scandium.
31. Correct option: (A)
According to Mendeleev's periodic table Eka Aluminium is later named as Gallium.
32. Correct option: (C)
According to Mendeleev's periodic table Eka –Silicon is later named as Germanium.
33. Correct option: (C)
According to Mendeleev's Periodic Law the properties of elements are the periodic function of their relative atomic masses. In this periodic table there are 7 periods and 8 groups.

LEVEL-II

1. Correct option: (C)

$$\text{Arithmetic mean} = \frac{\left(\textit{Atomic weight of Calcium} + \textit{Atomic weight of Barium}\right)}{2}$$

$$\text{Arithmetic mean} = \frac{\left(40.078 + 137.327\right)}{2} = \frac{177.405}{2} = 88.703$$

The atomic weight of strontium is approximately 87.62.
The pair with an arithmetic mean closest to the atomic weight of strontium is option C) Calcium and Barium with an arithmetic mean of 88.703, which is the closest to 87.62.
2. Correct option: (A)
Relation between all properties of an element.
3. Correct option: (D)
Elements which exhibit both metallic and non-metallic characters are called metalloids. Metalloids are located along the "staircase" in the periodic table, which separates metals from non-metals. They exhibit properties that are intermediate between those of metals and non-metals. Some common examples of metalloids include Silicon, Germanium and Arsenic.
(ii) Answer: Correct option: (D)
The arrangement of elements into groups of three is called the "triad" system of classification. In this system, elements with similar properties were grouped into sets of three based on their atomic weights. The triad system was one of the early attempts at organizing the elements and identifying patterns in their properties.
4. Correct option: (A)
In a Dobereiner triad, the atomic weight of the middle element is approximately equal to the arithmetic mean of the atomic weights of the other two elements.
Given that the atomic weight of 'C' (Carbon) is 40 and the atomic weight of 'E' is 137, we can find the atomic weight of 'H' (Hydrogen) using the arithmetic mean:

$$\text{Atomic weight of 'H'} = \frac{\left(\textit{Atomic weight of 'C'} + \textit{Atomic weight of 'E'}\right)}{2}$$

$$\text{Atomic weight of 'H'} = \frac{\left(40 + 137\right)}{2}$$

Atomic weight of 'H' = $\frac{177}{2}$

Atomic weight of 'H' = 88.5

Therefore, the atomic weight of 'H' is 88.5.

5. Correct option: (D)

A) The law of octaves was the first logical attempt to classify elements on the basis of atomic weights. Newland observed that every eighth element in the list had properties similar to the first, which he called the "law of octaves."

B) Periodicity of elements was recognized for the first time. Newland's work led to the recognition of periodicity in the properties of elements, indicating that elements with similar properties recurred at regular intervals.

So, both statements A and B are correct.

6. Correct option: (C)

The law of triads was proposed by Döbereiner, who noticed that certain groups of three elements had similar chemical properties, and the atomic weight of the middle element was the average of the other two elements. In this case, the elements Chlorine, Bromine, and Iodine form a triad according to this law:

Atomic weight of Bromine Br=

$$\frac{\left(Atomic\ weight\ of\ Chlorine\ \left(Cl\right)\ +\ Atomic\ weight\ of\ Iodine\ \left(I\right)\right)}{2}$$

$$Br=\ \frac{\left(35.453\ \left(Cl\right)\ +\ 126.904\ \left(I\right)\right)}{2}$$

$$Br\approx\ \frac{\left(35.453\ +\ 126.904\right)}{2}$$

$$Br\approx\ \frac{162.357}{2}$$

$$Br\approx 81.1785$$

The atomic weight of Bromine Br is approximately 81.1785, which is close to its actual atomic weight.

7. Correct option: (D)

According to Dobereiner's law of triads, chemically similar elements are arranged in the order of their atomic weights in groups of three elements.

Dobereiner observed that certain groups of three elements with similar chemical properties existed, and the atomic weight of the middle element was approximately the average of the other two elements in the triad. This implied that elements with similar chemical behaviors were found to have a relationship in terms of their atomic weights.

8. Correct option: (B)

Mendeleev's periodic cannot explain the existence of isotopes. Isotopes were discovered later, after Mendeleev's time, by J.J. Thomson (in 1912) and Frederick Soddy (in 1913). Isotopes are defined as the atoms of the same element that have the same number of protons but different numbers of neutrons, resulting in different atomic masses. The concept of isotopes was a significant advancement in atomic theory but was beyond the knowledge available during Mendeleev's era.

9. Correct option: (A)

A) Argon (Ar) – Potassium (K)

In the early versions of the periodic table, the atomic weight of (Ar) was placed before potassium (K), despite having a higher atomic weight. This was an anomaly that

Mendeleev faced when organizing the elements based on their properties and atomic weights. However, it was later discovered that the correct order should be as it is in the modern periodic table, with potassium (K) having a higher atomic weight than argon (Ar).

10. Correct option: (A)

According to Newland's law of octaves, the elements having similar properties are arranged in groups of eight, where the eighth element shows properties similar to the first. The correct option from the given choices is:

A) Lithium, Sodium, Potassium

11. Correct options: (A, B, D)

The correct statement(s) are:

A) Law of octaves could be best applied only up to the element Calcium.

B) Newly discovered elements could not fit into the octave structure.

D) Law of octaves was based on atomic weights.

A) Law of octaves could be best applied only up to the element Calcium: This statement is correct.

Newland's law of octaves was based on arranging elements in order of increasing atomic weights. He observed that elements with similar properties recurred every eighth element, similar to the notes in an octave of music. However, the pattern broke down after calcium, and the law was not applicable beyond that point.

B) Newly discovered elements could not fit into the octave structure: This statement is correct.

Newland's law of octaves was based on the existing elements known during his time. As new elements were discovered beyond calcium, they did not fit into the repeating pattern of the octaves, which was one of the limitations of his classification.

D) Law of octaves was based on atomic weights: This statement is correct.

Newland's law of octaves was primarily based on arranging elements in the order of their increasing atomic weights. He noticed that elements with similar properties occurred at regular intervals, much like the repeating pattern of musical octaves.

12. Correct option: (B, C and D)

Group VIII consists of 9 transition elements and they are further arranged into three triads based on similarities in their physical and chemical properties.

These triads are

- Iron, Cobalt, and Nickel
- Ruthenium, Rhodium, and Palladium
- Osmium. Iridium and Platinum

13. Correct options (A, C, D)

Lother Mayer curves represent the periodic variation of properties of elements with increasing atomic numbers. The peaks in these curves represent the elements with maximum values of a particular property. To identify the elements that are not at the peak of the Lother Mayer curves, we need to check the periodic properties of the given elements:

A) V (Vanadium): Vanadium is not at the peak of Lother Mayer curves. It is located in the middle of the d-block transition metals in the periodic table.

C) Al (Aluminum): Aluminum is not at the peak of Lother Mayer curves. It is a p-block element located in Group 13 of the periodic table.

D) Zn (Zinc): Zinc is not at the peak of Lother Mayer curves. It belongs to period 4.

14. Correct options: (A, C)

The correct statement is:

A) Sodium is a strongly electropositive metal.

C) Chlorine is a strongly electronegative non-metal.

Explanation:
A) Sodium is a strongly electropositive metal: This statement is correct. Sodium (Na) is an alkali metal located in Group I of the periodic table. Metals in Group I are highly electropositive, meaning they readily lose electrons to form positively charged ions (cations).
C) Chlorine is a strongly electronegative non-metal: This statement is correct. Chlorine is a non-metal and is one of the most electronegative elements on the periodic table. It readily gains electrons to form negatively charged ions (anions) in chemical reactions.
So, the correct statements are Sodium is a strongly electropositive metal, and Chlorine is a strongly electronegative non-metal.

15. Correct option: (B)

The classification proposed by Dobereiner, known as Dobereiner's law of triads, was based on the atomic masses of elements. He observed that certain groups of three elements had similar chemical properties, and the atomic weight of the middle element was approximately the average of the other two elements in the triad. This implied that elements with similar chemical behaviors were found to have a relationship in terms of their atomic masses. However, this classification was later replaced by more comprehensive periodic table arrangements, such as Mendeleev's periodic table, which were based on atomic weights but also considered other properties and trends.

16. Correct option: (B)

Metals are the elements that are malleable, ductile, and good conductors of heat and electricity. Let's break down each of these characteristics:
Malleability: Metals have the ability to be pressed into thin sheets without breaking. This property makes them suitable for various applications, such as metalworking and creating thin metal foils.
Ductility: Metals can be drawn into thin wires without breaking. This property is essential in the production of wires for electrical applications and other purposes.
Good conductors of heat and electricity: Metals have high electrical and thermal conductivity, allowing them to efficiently transfer heat and electricity. This characteristic makes metals indispensable for electrical wiring, electronics, and various heat transfer application.

17. Correct option: (A)

The elements which are brittle, bad conductors of heat and electricity are named as non-metals. Non-metals typically have opposite characteristics to metals:
Brittleness: Non-metals are generally brittle, meaning they are prone to breaking when subjected to stress. Unlike metals, which are malleable, non-metals do not easily deform without breaking.
Poor conductors of heat and electricity: Non-metals have low electrical and thermal conductivity compared to metals. They do not allow efficient transfer of heat and electricity, which makes them unsuitable for electrical and heat conduction applications.

18. Correct option: (C)

The elements that possess both metallic and non-metallic characteristics are named as metalloids. Metalloids are a group of elements that lie along the boundary between metals and non-metals in the periodic table. They exhibit properties of both metals and non-metals, making them unique in their behavior.
Some of the key characteristics of metalloids include:
Intermediate electrical conductivity: Metalloids have electrical conductivity that falls between that of metals and non-metals. Their conductivity can be altered by factors such as temperature and impurities.

Some ductility and malleability: Metalloids may exhibit limited ductility and malleability, but not to the extent of metals.
Brittle in solid form: Metalloids are generally brittle in their solid form, especially at room temperature.
Semiconducting behavior: Many metalloids, such as silicon and germanium, are crucial components in semiconductor devices used in electronics.
Examples of metalloids include Silicon, Germanium, Arsenic, Antimony, and Tellurium.

19. Correct Matching is: a-R; b-S; c-P; d-T

a) Correct option is R.
Antoine Lavoisier, a French chemist often referred to as the "father of modern chemistry," made significant contribution on classification of elements, including metals and non-metals. In the late 18th century, Lavoisier proposed a classification of elements based on their chemical properties.
b) Correct option is S.
One of the significant triads Dobereiner observed in his classification was that of calcium, strontium, and barium. These three elements have similar chemical properties, and their atomic weights form a triad relationship:
Calcium (Ca) - Atomic weight approximately 40
Strontium (Sr) - Atomic weight approximately 87.5
Barium (Ba) - Atomic weight approximately 137
c) Correct option is P.
Lothar Meyer is a German chemist plotted a graph between atomic weight and atomic volume and he find out that elements with similar properties occupied the similar positions on the graph.
d) Correct option is T.

We can see that the atomic weight of strontium ($87.62\ \frac{g}{mol}$) is approximately the average

of the atomic weights of calcium ($40.08\ \frac{g}{mol}$) and barium ($137.33\ \frac{g}{mol}$)

$$= \frac{\left(40.08\ g/mol\ +\ 137.33\ g/mol\right)}{2} \approx 87.71\ \frac{g}{mol}$$

20. Correct Matching is: a-Q; b-R; c-S; d-P

a) Correct option is Q
Chlorine is the eighth element after fluorine. The chemical properties of fluorine and chlorine are similar they are halogens belongs to 7th period in periodic table.
b) Correct option is R.
According to Newland's classification of elements, the properties of sulphur are similar to those of oxygen because sulphur is the eighth element starting from oxygen.
c) Correct option is S.
According to Newland's classification of elements, the properties of Aluminium are similar to those of Boron because sulphur is the eighth element starting from oxygen.
d) Correct option is P.
According to Newland's classification of elements, the properties of Phosphorous are similar to those of Nitrogen because sulphur is the eighth element starting from oxygen.

21. **Correct Matching is: a-R; b-P; c-S; d-Q**

a) Correct option is R.

Hydrogen belongs to the first period of the periodic table. In the modern periodic table, elements are arranged in periods based on their increasing atomic numbers. Hydrogen has an atomic number of 1, which is the smallest and the first in the periodic table.

b) Correct option is P.

Lithium belongs to the second period of the periodic table. In the modern periodic table, elements are arranged in periods based on their increasing atomic numbers. Lithium has an atomic number of 3, which places it in the second period.

c) Correct option is S.

Sodium belongs to the third period of the periodic table. In the modern periodic table, elements are arranged in periods based on their increasing atomic numbers. Sodium has an atomic number of 11, which places it in the third period.

d) Correct option is Q.

Potassium belongs to the fourth period of the periodic table. In the modern periodic table, elements are arranged in periods based on their increasing atomic numbers. Potassium has an atomic number of 19, which places it in the fourth period.

22. **Correct Answer is: 3**

The number of elements present in a triad is three. In the early development of the periodic table, scientists noticed that certain groups of three elements have similar chemical properties. These groups were called "triads."

23. Correct Answer is: 8

According to Newlands' idea, when elements were arranged in order of increasing atomic weights, every eighth element exhibited similar properties to the first, forming an "octave."

24. Correct Answer is: 8

The number of groups in the modified Mendeleev's periodic table would be the same as the number of groups in the modern periodic table. In the original version of the periodic table proposed by Dmitri Mendeleev, there were eight groups, which were organized based on similar chemical properties.

3. Solutions

Solutions

LEVEL-I

1. Correct option: (C)

A homogeneous mixture of two or more components is called a solution. In a solution, the compounds are evenly distributed at the molecular or atomic level, resulting in a uniform appearance and properties throughout the mixture.

2. Correct option: (A)

If a solution consists of two components, it is called a binary solution. In a binary solution, there are two different substances dissolved in each other, creating a homogeneous mixture with a uniform composition.

3. Correct option: (A)

The substance which is relatively less in quantity in a solution is called the solute. The solute is present in a smaller amount compared to the solvent in the solution.

4. Correct option: (C)

If a solution is composed of three components, it is called a ternary solution. The term "ternary" indicates the presence of three different components dissolved in each other to form a homogeneous mixture with a uniform composition.

5. Correct option: (B)

There is only one phase present in a true solution. A true solution is a homogeneous mixture where the solute is uniformly dispersed in the solvent. Since the solute particles are distributed evenly at the molecular level, there is no visible distinction between the solute and solvent, resulting in a single-phase system.

6. Correct option: (D)

All solutions are characterized by all of the following:

A) **Absence of settling**: Solutions are homogeneous mixtures in which the solute particles are evenly distributed throughout the solvent, and there is no settling or separation of components over time.

B) **Molecular or ionic state of subdivision of the components**: In a solution, the solute is present in molecular or ionic form, that is the particles are individually dispersed.

C) **Homogeneity**: Solutions are uniform in composition throughout, meaning they have the same properties and composition at every point within the mixture. There are no visible differences or phase boundaries between the components.

7. Correct option: (B)

Solution of water and sugar is commonly known as syrup.

8. Correct option: (C)

The solute present in seltzer solution is carbon dioxide (CO_2). Seltzer is a type of carbonated water, and the carbon dioxide gas dissolved in the water that gives it its fizziness.

9. Correct option: (A)

Solution of water and salt is called brine. Brine is commonly used for pickling and preserving food, as well as in various industrial processes.

10. Correct option: (B)

An alloy is an example of a solid-solid type solution. Alloys are homogeneous mixtures of two or more elements at least one of the components is a metal. The atoms of the different elements mix together forming a solid solution. Common examples of alloys include brass (copper and zinc), stainless steel (iron and chromium), and bronze (copper and tin).

11. Correct option: (C)

Soda water is an example of a gas in liquid type of solution. It is a mixture of carbon dioxide (gas) dissolved in water (liquid). The carbon dioxide gives soda water its characteristic fizziness and is present as a gas dissolved in the liquid.

12. Correct option: (A)

If air is taken as a binary solution, the solvent is considered to be nitrogen (N_2). Air is primarily composed of nitrogen (approximately 78% by volume) and oxygen (approximately 21% by volume), along with trace amounts of other gases such as carbon dioxide, argon, and others. In this context, nitrogen (N_2) is considered the solvent, and oxygen (O_2) along with the other trace gases are considered the solutes in the binary solution.

13. Correct option: (C)

Oil in water

A heterogeneous mixture is a mixture in which the components are not uniformly distributed and can be distinguished as separate phases within the mixture. Among the options given, "Oil in water" is a heterogeneous mixture. When oil is mixed with water, the two substances do not fully dissolve into each other, and oil droplets remain separate from the water, creating visible distinct separate phases.

14. Correct option: (C)

If the quantity of solute is more in a solution, then the solution is said to be a concentrated solution. In a concentrated solution, a relatively large amount of solute is dissolved in a given quantity of solvent. The concentration of the solute is higher compared to other types of solutions.

15. Correct option: (B)

A solution in which alcohol is used as the solvent is known as an "Alcoholic" solution. In this the term "alcoholic" refers to a solution where alcohol serves as the medium for dissolving one or more solutes. These solutions are common in various industries, for example, alcoholic beverages like wine, beer, and spirits are examples of alcoholic solutions where alcohol is the solvent.

16. Correct option: (C)

A solution that cannot dissolve any more solute at a given temperature is known as a "Saturated solution." In a saturated solution, the maximum amount of solute has been dissolved in the solvent and any additional solute added will not dissolve but will remain undissolved at the bottom of the container.

17. Correct option: (B)

A solution that can dissolve some more solute at a given temperature is known as an "**Unsaturated solution**." In an unsaturated solution, the amount of solute dissolved in the solvent is less than the maximum amount that could be dissolved at that temperature.

18. Correct option: (A)

A solution that has excess of solute dissolved in it than required for its saturation at a given temperature is known as a "Super saturated solution." That means more solute has been dissolved in the solvent than required under that specific temperature and conditions. Super saturated solutions are often formed by dissolving the solute at a higher temperature and then cooling the solution slowly without allowing the excess solute to precipitate out. As a result, the solution becomes unstable and can crystallize if disturbed, releasing the excess solute and returning to a saturated state.

19. Correct option: (A)

Milk is actually a colloidal suspension, not a true solution. It is composed of water as the solvent and various particles, including proteins, fats, and minerals, as the solute. These particles are dispersed throughout the liquid but do not dissolve completely.

20. Correct option: (D)

Cough syrup contains solid particles or droplets of active ingredients that do not dissolve completely in the liquid. These solid particles remain suspended in the liquid, and over time, they may settle down at the bottom. To use cough syrup, you usually need to shake the bottle to disperse the solid particles before taking the medicine.

21. Correct option: (A)

The phenomenon of scattering of visible light by the particles of a colloid is known as the Tyndall effect. When a beam of light passes through a colloid, it gets scattered by the dispersed particles, making the path of the beam visible. This effect is not observed in true solutions, where the particles are too small to scatter light significantly.

22. Correct option: (B)

A colloid is a type of mixture in which the size of the solute particles is within the range of approximately 1 nanometer to 100 nanometers. In colloids, the particles are larger than those found in true solutions but smaller than those in suspensions.

23. Correct option: (C)

Salt and water

The Tyndall effect is defined as scattering of light by the particles in a colloid or suspension. It is observed when the particles in the mixture are large enough to scatter light. Among the options provided, salt and water form a true solution because the salt particles dissolve completely in water, and the resulting mixture is homogeneous.

24. Correct option: (B)

The solution in which the size of the solute particles is greater than 100 nanometers is called a colloid. In colloids, the particle size falls in the range of approximately 1 nm to 1000 nm, and these particles are large enough to scatter light, resulting in the Tyndall effect.

25. Correct option: (B)

A heterogeneous mixture is defined as components are not uniformly distributed throughout the mixture. Milk is a heterogeneous mixture because it consists of various components like water, fats, proteins, vitamins, and minerals, which are not evenly dispersed throughout the liquid.

LEVEL-II

1. Correct option: (C)

The correct statement is C) Solutions are homogeneous mixtures.

A solution is defined as a type of mixture in which the solute is uniformly distributed and dissolved in another substance. The particles of the solute mix completely with the particles of the solvent, creating a uniform composition throughout the mixture. Because of this uniform distribution, solutions are considered homogeneous mixture.

2. Correct option: (B)

Camphor in air is an example of a solid (camphor) dispersed in a gas (air). When you place a piece of camphor in the open air, it gradually evaporates and forms a gaseous phase around it, which you can observe as peculiar camphor smell. The solid camphor particles disperse into the air, making it an example of a solid in a gas mixture.

3. Correct option: (C)

Dissolution of glucose in water is correct answer among the given options.

A physical change is defined as a change in physical properties of a substance without disturbing its chemical properties. When glucose is added to water, it gets uniformly distributed in the water, forming a homogeneous mixture. However, the chemical composition of glucose remains the same; it does not undergo any chemical reaction during the dissolution process.

On the other hand, other options involve burning or combustion processes, which are chemical changes. During these processes, the substances undergo chemical reactions and are converted into different compounds.

4. Correct option: (A)

Gas in solid is correct option among all.

Occlusion of hydrogen on palladium is an example of gas (hydrogen) being trapped within the solid structure of palladium. This process is known as "occlusion," and it represents a gas (hydrogen) being incorporated into the lattice structure of the solid (palladium) without forming a chemical compound.

5. Correct option: (C)

Ethanol dissolved in water is the correct option among all.

A liquid-in-liquid type solution, also known as a homogeneous mixture, is a solution where one liquid is uniformly dispersed and dissolved in another liquid. Among the options provided, ethanol dissolved in water is a liquid-in-liquid type solution. When ethanol (a liquid) is mixed with water (liquid), they form a homogeneous mixture with no visible boundaries between the two components.

6. Correct option: (B)

The mixture of benzene and toluene is an example of a liquid-in-liquid type solution. When benzene and toluene are mixed together, they form a homogeneous mixture, with one liquid (benzene) dissolved and uniformly dispersed in the other liquid (toluene).

7. Correct option: (C)

A homogeneous mixture is defined as a mixture in which the components are uniformly distributed and cannot be visibly distinguished. Among the options provided, "Salt in water" is homogeneous in nature. When salt is dissolved in water, it completely dissolves and forms a homogeneous solution, and you cannot see separate salt particles within the water. The salt molecules disperse evenly throughout the water, resulting in a uniform mixture.

8. Correct option: (C)

A heterogeneous mixture is defined as a mixture in which the components are not uniformly distributed, and the different substances can be visibly distinguished. Among the options provided, "Oil in water" is a heterogeneous mixture. When oil is added to water, it does not dissolve but forms separate droplets that can be seen floating on the surface or dispersed throughout the water.

9. Correct option: (B)

Option B is false. Solutions are not pure substances; they are homogeneous mixtures. A pure substance consists of only one type of particle, whereas a solution is a mixture of two or more substances, where one substance (the solute) is dissolved uniformly in another substance (the solvent).

Options A, C, and D are true:

A) **Solutions are homogeneous mixtures**: A homogeneous mixture is a mixture in which the components are uniformly distributed and cannot be visibly distinguished. In solutions solute is uniformly distributed in the solvent, resulting in a uniform composition throughout the mixture.

C) **The solute is uniformly distributed in the solvent**: This statement is true, as discussed above. In a solution, the solute particles are uniformly distributed in the solvent, ensuring a homogeneous mixture.

D) **Preparation of a solution is not a chemical change**: The process of preparing a solution involves dissolving a solute in a solvent, which is a physical change, not a chemical change. No new substances with different chemical properties are formed during the preparation of a solution.

10. Correct option: (C)

C) Sugar dissolved in water - solid in water

Option C is correctly matched. When sugar (a solid) is dissolved in water, it forms a homogeneous mixture, where the sugar particles become uniformly distributed in the water, resulting in a solid-in-liquid type solution. The sugar particles do not settle at the bottom and cannot be visually distinguished from the water, making it a solid dissolved in a liquid.

11. Correct options: (A, B, C, D)

A) **A solution in which alcohol is used as solvent is called an alcoholic solution** - This statement is correct. When alcohol is used as the solvent for other solute which are dissolved in it, the resulting solution is called an alcoholic solution. For example, if you dissolve sugar in alcohol, you get an alcoholic solution of sugar.

B) **A solution that cannot dissolve any more solute at a given temperature is known as a saturated solution** - This statement is correct. A saturated solution is one where the maximum amount of solute that can be dissolved in the solvent has already been dissolved at a specific temperature. Any additional solute added will not dissolve and will remain as undissolved particles.

C) **A solution that can dissolve some more solute at a given temperature is known as an unsaturated solution** - This statement is correct. An unsaturated solution is one where more solute can be dissolved in the solvent at a specific temperature. It means the solution has not reached its maximum capacity to hold more solute.

D) **All solutions are mixtures** - This statement is correct. A solution is a type of mixture where two or more substances are mixed together, but they do not chemically react with each other. Instead, they form a homogeneous mixture where the components are uniformly distributed throughout.

12. Correct options: (B, C and D)

Except option A all the given options are incorrect.

By using process filtration and decantation we will not be able to separate all the components in every conditions of the solution. We know that solution is a homogeneous mixture, and it cannot have any particular chemical formula.

13. Correct option: (A)

In brine solution, the solute is NaCl (sodium chloride). Brine is a solution of salt (NaCl) in water.

The solute (NaCl) dissolves in the solvent (water) to form a homogeneous mixture known as brine.

14. Correct option: (D)

Gas, gas - This is the correct option since air is a mixture of different gases, with no solid or liquid components involved.

15. Correct option: (C)

A solution is a homogeneous mixture of a solute dissolved in a solvent. Let's examine each option:

A) **Milk** - Milk is not a solution. It is an emulsion, which is a type of colloid. It consists of small droplets of fat and protein dispersed in water.

B) **Cheese** - Cheese is not a solution either. It is a solid product obtained from the coagulation of milk proteins, and it may contain fat and other components

C) **Sugar in water** - This is a solution. When sugar (solute) is added to water (solvent) and dissolves completely, it forms a homogeneous mixture called a sugar solution.

D) **Sawdust in water** - This is not a solution. Sawdust is insoluble in water, so it will not dissolve.

16. Correct option: (D)

Air is a mixture of different gases, such as nitrogen, oxygen, carbon dioxide, and trace amounts of other gases. Since all these gases are distributed well and mixed together, it forms a gas-gas solution.

17. Correct option: (C)

Solid solution → Camphor in air

The reason is that camphor in air does not form a solid solution. Camphor is a solid compound, and when it is placed in the air, it undergoes sublimation, means it transitions directly from a solid to a gaseous state without forming a solution with the air. Therefore, camphor in air is not a solid solution.

18. Correct option: (B)

Amalgam is a type of alloy where mercury is the solvent (liquid) and another metal, such as silver, gold, or tin, is the solute (solid). The metals are dissolved in the liquid mercury, forming a homogeneous mixture, which is a liquid in a solid state.

19. Correct Matching is: a-R, b-S, c-Q, d-P

a) Correct option is R.

A binary solution refers to a type of solution that contains two components: a solute and a solvent. The solute is the substance that gets dissolved in the solvent to form the solution. The solvent is the component in which the solute dissolves resulting a homogenous mixture.

b) Correct option is S.

The word “Ternary" indicates that there are three components present, and it doesn't specify the nature of these components or the exact type of mixture.

Ternary Solution: This is a homogeneous mixture where three substances are dissolved in a solvent. For example, a solution containing water as solvent and dissolved salts (sodium chloride and potassium nitrate) could be considered a ternary solution.

c) Correct option is Q.

It is defined as a homogeneous mixture where four substances are dissolved in a solvent. For example, a solution containing water as solvent and dissolved salts (sodium chloride, potassium nitrate, and calcium sulfate) could be considered a quaternary solution.

d) Correct option is P.

In a dilute solution, the amount of solute (the substance being dissolved) is relatively small compared to the amount of solvent (the substance doing the dissolving). Dilution occurs when additional solvent is added to a solution, reducing the concentration of the solute. This can be achieved by adding more solvent to an existing solution or by mixing a concentrated solution with more solvent.

20. Correct Matching is: a-Q, b-S, c-P, d-R

a) Correct option is Q.

A solution of water and sugar is commonly named as a "sugar solution" or a "sugar-water solution." In this solution, water is the solvent (the substance that does the dissolving), and sugar is the solute (the substance being dissolved).

b) Correct option is S.

A water and salt solution is commonly called a "Brine" or "salt solution."

When salt (example NaCl) dissolves in water, it forms a homogeneous mixture where the water acts as the solvent and the salt as the solute. The resulting solution is formed.

c) Correct option is P.

It is the solution made by water and carbon dioxide, resulting in a bubbly and effervescent beverage. The carbonation process involves dissolving carbon dioxide gas into the water, creating carbonic acid (H_2CO_3).

d) Correct option is R.

When ammonia gas is dissolved in water, the resulting solution formed is called "ammonia solution" or "aqueous ammonia." This solution is commonly used in various applications, such as cleaning, fertilizers, and as a source of ammonia in chemical reactions.

$$NH_4OH(aq) \rightleftharpoons NH_4(aq) + OH_(aq)$$

21. Correct Answer is: 6 types

Based on the physical state of the solute and solvent present in a solution, solutions are classified into six types:

Gas in Gas (gaseous solution)
Gas in Liquid (liquid-gas solution)
Gas in Solid (solid-gas solution)
Liquid in Liquid (liquid-liquid solution)
Liquid in Solid (solid-liquid solution)
Solid in Solid (solid-solid solution)

22. Correct Answer is: 4

The number of components in a quaternary solution is four.

23. Correct Answer is: one phase.

In a true solution, there is only one phase present. A true solution is defined as a homogeneous mixture of two or more substances where the solute particles are uniformly distributed and individual solute particles cannot be observed with the naked eye or through simple filtration.

24. Correct Answer is: 3 types.

Depending on the size of the solute particles, solutions can be classified into three main types:

True Solutions
Colloidal Solutions
Suspension Solutions

25. Correct Answer is: 4 types.

Dilute Solutions
Concentrated Solutions
Saturated Solutions
Supersaturated Solutions

4. Soil
Solutions

LEVEL-I

1. Correct option: (C)Minerals, organic matter, air and water
In addition to the rock particles, the soil contains **minerals, organic matter, air and water.**

2. Correct option: (B) Clayey soil
The water holding capacity is the highest in **clayey soil. Clay particles** have the ability to physically and chemically "hold" water molecules to the particle more tightly than sands or silts.

3. Correct option: (C)Clay
The largest particle is gravel, and the smallest one is **clay**. The particles that makeup soil are categorized into three groups by size – sand, silt, and clay. Single clay particles cannot be seen with a naked eye. Size is less than 0.002 mm in diameter.

4. Correct option: (A)Low percolation rate of water
The percolation rate of **clay soil** is very low which means that very less water seeps through the soil. This makes it most suitable for the cultivation of paddy.

5. Correct option: (B) Erosion
The carrying away of the upper fertile layer of the soil by wind or water is called **soil erosion**.

6. Correct option: (C) 2 and 4
The manufacturing of paper and plywood materials on a large scale is a cause of soil erosion This is because it involves the process of deforestation.

7. Correct option: (C) Flood
Flooding cannot be an effect of soil erosion.

8. Correct option: (D) Sand
Sand is the component of soil that makes the air above the soil shimmer on a hot summer day. Due to the extreme water vapor present on the soil, the air above it appears shimmering on a hot day.

9. Correct option: (B) Deforestation
Deforestation can not prevent soil erosion .

10. Correct option: (D) Weathering
The process by which huge rocks are broken down into small particles by the action of sun's heat , wind, rain and flowing water, etc. is called weathering**.**

11. Correct option: (C) 1 and 3
Jute bags and cardboards do not cause soil pollution. Jute's most significant characteristic is that it is biodegradable.

12. Correct option: (C) Humus
Humus is **a** substance made from dead leaves and plants, that put into the ground to help plants grow. Humus is partially decayed organic matter. Function of humus: It makes soil porous, thereby increasing its air and water holding capacity.

13. Correct option: (D) Sandy-loam soil
Almost two-thirds of the agricultural land in the south-west region has a duplex soil profile with **sandy loam surface soils** over sandy clay subsoils. These soils are susceptible to waterlogging when the amount of rainfall exceeds the ability of the soil to drain away soil moisture**.**

14. Correct option: (B) Silty soil
Silt is not a soil type, it is the clay, fine sand carried by a river. It is deposited as sediments usually on harbours and river banks. Soil contains differently sized particles namely clay,silt and sand.
15. Correct option: (A) Clayey soil
Wheat is grown in a variety of soils of India. Soils with a **clay loam or loam texture, good structure and moderate water holding capacity** are ideal for wheat cultivation.
16. Correct option: (A) 10%
17. Correct option: (D) 50 percent
This water was added dropwise to 120 g of dry soil kept on filter paper in a funnel. When the water just started dripping from the funnel, the amount of water left in the measuring cylinder was found to be 140 mL.
18. Correct option: (D)4
Soils are named and classified based on their horizons. The soil profile has **four distinct layers**: 1) O horizon; 2) A horizon; 3) B horizon, or subsoil; and 4) C horizon, or soil base . The O horizon has freshly decomposing organic matter—humus—at its surface, with decomposed vegetation at its base.
19. Correct option: (A) Humus
Humus is the dead and decaying bits of animals and plants present in the **topmost layer of soil**.
20. Correct option: (B) Topsoil
The A-Horizon is also known as **the Topsoil .**This layer is also rich in organic material and commonly known as the humus layer. This majorly consists of both organic matter and other decomposed type materials.

21 - 55 questions Key

21)	**C**	**22)**	**B**	**23)**	**C**	**24)**	**A**	**25)**	**D**	**26)**	**C**	**27)**	**D**
28)	**A**	**29)**	**B**	**30)**	**D**	**31)**	**B**	**32)**	**C**	**33)**	**B**	**34)**	**A**
35)	**C**	**36)**	**B**	**37)**	**B**	**38)**	**C**	**39)**	**A**	**40)**	**A**	**41)**	**B**
42)	**C**	**43)**	**A**	**44)**	**C**	**45)**	**A**	**46)**	**A**	**47)**	**D**	**48)**	**D**
49)	**C**	**50)**	**B**	**51)**	**D**	**52)**	**C**	**53)**	**B**	**54)**	**D**	**55)**	**B**

LEVEL-II

1. Correct option: (B) Topsoil
Topsoil- This is the layer that we call "topsoil" and it is located just below the O Horizon. This layer is made up of minerals and decomposed organic matter and it is also very dark in color. This is the layer that many plants roots grow in.
2. Correct option: (A) A-Horizon
Humus is the dead and decaying bits of animals and plants present in the topmost layer of soil.
3. Correct option: (C) Humus
Humus is a substance made from dead leaves and plants, that put into the ground to help plants grow. Humus is partially decayed organic matter. Function of humus: It makes soil porous, thereby increasing its air and water holding capacity.
4. Correct option: (B) Topsoil
Plant seeds for plants to grow in **the topsoil**. It is the topmost layer of soil. Plants get

minerals and organic matter or humus required for their growth from this layer.

5. Correct option: (A) A-Horizon

The topsoil or A-Horizon is soft and porous to hold enough air and water. In this layer, the seed germination takes place and new roots are produced which grows into a new plant. This layer consists of microorganisms such as earthworms, fungi, bacteria, etc.

6. Correct option: (B) B-Horizon

The B horizon or subsoil is where soluble minerals and clays accumulate. This layer is lighter brown and holds more water than the topsoil because of the presence of iron and clay minerals. There is less organic material.

7. Correct option: (C) Subsoil

B-Horizon is also known as subsoil.

8. Correct option: (B) False

C-horizon is a layer in which the soils have developed. Topsoil (A-Horizon) is darker than subsoil (B-Horizon).

9. Correct option: (A) TRUE

Subsoils are generally lower in nutrients than the surface layer. This can reduce yield in dryland regions, where continued root growth and function are essential to enable crops to extract water from the subsoil once the topsoil is dried or available nutrients used up by the crop.

10. Correct option: (B) NO

Humus is the dead and decaying bits of animals and plants present in the **topmost layer of soil**.

11. Correct option: (B) FALSE

Subsoil or the B horizon, is the middle soil layer. It has **fewer organisms** and less organic materials than both the A and O horizons.

12. Correct option: (C) C-Horizon

The layer of soil just below the subsoil is called Sub-stratum. Sub-stratum is also called C-horizon.

13. Correct option: (D) Substratum

The layer of soil just below the subsoil is called Sub-stratum. Sub-stratum is also called C-horizon

14. Correct option: (C) Parent rock

Below the C-horizon, we have an unweathered solid rock called the parent rock .

15 - 20 questions Key

15) C 16) D 17) AD 18) ABCD 19) AC
20) a-R, b-P, c-Q,d-S

5. Chemistry in Daily Life
Solutions

LEVEL 1

1. **Correct option: (D)**
Crook's glass (cerium oxide) which absorbs high UV light and is used to prepare spectacles.
2. **Correct option: (D)**
The basic components of paint include:
(a) Pigment: Provides colour and opacity to the paint.
(b) Vehicle (or binder): Carries the pigment and binds it to the painted surface.
(c) Solvent: A liquid that helps to adjust the consistency of the paint and evaporates after application.
3. **Correct option: (C)**
Vehicle tires are typically made of rubber, which is a polymer. Rubber is composed of long chains of repeating molecular units, making it a polymer. The other options—cement, glucose, and common salt—do not have polymers in their basic composition.
4. Correct option: (D)
Among the options given, bromine (Br_2) is present in a liquid state at room temperature. The others fluorine (F_2), hydrogen (H_2), and chlorine (Cl_2) are gases at room temperature.
5. Correct option: (A)
Graphite is a good conductor of electricity primarily due to the presence of free electrons. In the structure of graphite, carbon atoms are arranged in hexagonal layers, and each carbon atom is bonded to three others in a trigonal planar arrangement. The fourth outer electron of each carbon atom is free to move within the layers, forming a sea of delocalized electrons. These free electrons are able to carry an electric current, making graphite a good conductor.
6. Correct option: (D)
Duralumin is an aluminium alloy that typically contains aluminium, copper, and small amounts of other elements.
7. Correct option: (C)
Alloyed gold often contains silver or copper, or a combination of both. These metals are added to gold to create alloys with desirable properties, such as increased hardness or changes in colour.
8. Correct option: (A)
Bronze is generally more resistant to corrosion than brass. Bronze is an alloy of copper and tin, and its corrosion resistance can be attributed to the formation of a protective layer of copper oxide on its surface. On the other hand, brass is an alloy of copper and zinc and may be more susceptible to corrosion, especially in certain environments.
9. **Correct option: (C)**
Hydrochloric acid (HCl) is commonly used in the laboratory for the preparation of silver chloride (AgCl) by reacting it with silver nitrate (AgNO3). The reaction is as follows:
$HCl + AgNO_3 \rightarrow AgCl + HNO_3$
10. **Correct option: (D)**
Common salt refers to sodium chloride (NaCl), and it does not contain calcium chloride (CaCl2), silver chloride (AgCl), or a combination of NaCl, AgCl, and NaI as mentioned in the other options. The chemical formula for common salt is NaCl.
11. **Correct option: (D)**
Blue vitriol, also known as copper sulphate or $CuSO_4$, gets its blue colour from the presence of copper ions (Cu^{+2}) in the compound. When copper sulphate is dissolved in water, it forms

a blue solution due to the hydrated copper ions present in the solution. The blue colour is not from water of crystallization, but rather from the copper ions.

12. **Correct option: (A)**

Calcium oxide, represented as CaO, is commonly known as lime. It is produced by heating calcium carbonate ($CaCO_3$) to high temperatures, causing it to undergo thermal decomposition and release carbon dioxide, leaving behind calcium oxide. Lime is used in various applications, including as a construction material and in agriculture.

13. Correct option: (B)

When zincite is strongly heated, it undergoes thermal decomposition to form zinc oxide (ZnO). This process involves the breakdown of zincite into its constituent elements, releasing oxygen and leaving behind zinc oxide.

14. **Correct option: (D)**

Stainless steel is an alloy that typically contains iron (Fe), chromium (Cr), nickel (Ni), and other elements. The combination of these elements imparts corrosion resistance, strength, and other desirable properties to stainless steel. Therefore, all the elements mentioned in the options iron, chromium, and nickel are commonly present in stainless steel.

15. Correct option: (C)

The major component of biogas is methane (CH_4). Biogas is a renewable energy source produced through the anaerobic digestion or fermentation of organic matter such as animal waste, agricultural residues, and other organic materials. Methane is the primary combustible component in biogas and is used as a clean-burning fuel for various applications.

16. Correct option: (A)

Portland cement is typically made by combining limestone and clay in specific proportions and then heating the mixture in a kiln. The process of heating the mixture results in the formation of clinker, which is then ground into a fine powder to produce Portland cement. The correct option is the mixture of limestone and clay.

17. Correct option: (C)

In the dry process of cement manufacturing, the fine powder of limestone and clay is produced through the grinding process. The raw materials are first quarried and then crushed into smaller pieces before being finely ground into the powder form. Therefore, the correct term for this process is "grinding."

18. Correct option: (A)

Silica, also known as silicon dioxide (SiO_2), is a key component used in the making of glass. Silica is often derived from sand and provides the main structural component of glass. Other materials, such as soda ash and limestone, may also be added to the mix depending on the type of glass being produced.

19. **Correct option: (A)**

Feldspar is used in the making of glass. It is a common raw material in glass manufacturing, contributing to the composition of the glass batch along with silica (sand) and other materials. While silicone is related to silicon, it is not typically used directly as a primary raw material in glass making.

20. Correct option: (A)

The blowing of glass is commonly used in the manufacture of glassware. This technique involves shaping molten glass by blowing air into a gather of glass on the end of a blowpipe, allowing artisans to create various glass objects such as bottles, vases, and other decorative or functional items.

21. Correct option: (C)
A monomer is the fundamental unit or building block of a polymer. Monomers are small molecules that can undergo polymerization, a chemical reaction where they link together to form a larger molecule called a polymer. The process of combining monomers to form a polymer involves repeated chemical units, resulting in a macromolecule with a high molecular weight.

22. Correct option: (B)
The monomer of polyethylene is ethylene. Polyethylene is a polymer made by polymerizing ethylene molecules, which consist of two carbon atoms and four hydrogen atoms. The chemical formula of ethylene is C_2H_4 and through polymerization, it forms the repeating units of polyethylene.

23. Correct option: (D)
"Chain" is not a polymer; it is a term that can refer to a structure or arrangement but is not a specific polymer material. PVC (Polyvinyl chloride), plastic (which is a general term for various synthetic or semi-synthetic polymers), and Nylon are examples of polymers.

24. Correct option: (C)
Reusable plastics are often referred to as thermoplastics. Thermoplastics are a type of polymer that becomes mouldable when heated and solidifies upon cooling. This property allows them to be melted and reshaped multiple times, making them suitable for reuse. Examples of thermoplastics include polyethylene, polypropylene, and polyvinyl chloride (PVC).

25. Correct option: (D)
All the options—melamine, formaldehyde, and Bakelite—are examples of thermosetting plastics. Thermosetting plastics are polymers that, once moulded and set, cannot be melted or reshaped through heating. They undergo a chemical change during the curing or setting process, creating a strong and rigid structure. Melamine, formaldehyde (used in the production of phenolic resins), and Bakelite are common examples of thermosetting plastics.

26. Correct option: (C)
Starch is a carbohydrate. Carbohydrates are organic compounds made up of carbon, hydrogen, and oxygen, and they serve as a primary source of energy for living organisms. Starch is a complex carbohydrate consisting of many glucose units linked together. Lemon, butter, and salt, on the other hand, do not belong to the carbohydrate group.

27. Correct option: (D)
Keratin is a fibrous structural protein that is a major component in the structure of hair, nails, and the outer layer of the skin. It belongs to the protein category.

28. Correct option: (A)
Toilet soap typically contains better quality and more refined vegetable oils or fats compared to laundry soap. The quality of vegetable oil used in soap production affects the soap's texture, lather, and overall quality. Toilet soaps are designed for personal hygiene and often contain higher-quality ingredients to be gentler on the skin compared to laundry soaps, which are intended for cleaning fabrics.

LEVEL - II

1. Correct option: (C)
Zinc is a relatively soft metal with limited strength compared to other metals. Tensile strength is the maximum amount of tensile (pulling) stress that a material can withstand without breaking.

2. Correct option: (D)
Magnesium is not an alloy; it is a chemical element. The other options—brass, bronze, and magnalium—are alloys:
Brass is an alloy of copper and zinc.
Bronze is an alloy of copper and tin.
Magnalium is an alloy of aluminium and magnesium.

3. Correct option: (B)
Tinning refers to the process of coating something with a thin layer of tin. In this context, the element used for tinning is tin itself. The process is commonly used to coat iron or steel objects with a layer of tin to prevent corrosion.

4. Correct option: (B)
The galvanization process involves coating iron or steel with a layer of zinc to protect the metal from corrosion. Therefore, the two metals involved in the galvanization process are iron and zinc.

5. Correct option: (C)
Duralumin is an aluminium alloy that is commonly used in the manufacture of aircraft frames because it is lightweight and has high tensile strength, making it suitable for applications where strength and low weight are important, such as in aviation. The other options do not accurately describe the characteristics of duralumin.

6. Correct option: (B)
Helium is a noble gas. Noble gases are a group of elements that are generally inert and do not readily form compounds with other elements. Helium is chemically inert and belongs to the noble gas group in the periodic table.

7. Correct option: (D)
The elements typically present in stainless steel include iron as the base metal, chromium for corrosion resistance, carbon for strength, and nickel for additional corrosion resistance and other properties.

8. Correct option: (A)
The chemical composition of limestone is primarily calcium carbonate ($CaCO_3$). Limestone is a sedimentary rock composed mostly of the mineral calcite, which is a crystalline form of calcium carbonate.

9. Correct option: (C)
The chemical name of sand is silicon dioxide (SiO_2). Sand is composed of tiny grains of quartz, which is a crystalline form of silicon dioxide.

10. Correct option: (D)
Copper is the most suitable metal for telecommunication cables. It has excellent electrical conductivity, which makes it well-suited for transmitting electrical signals over long distances with minimal loss. Copper is widely used in the construction of electrical and telecommunication cables due to its conductivity, flexibility, and corrosion resistance.

11. Correct option: (D)
Distillation is not involved in the preparation of sodium chloride from seawater. The typical steps include dissolution of seawater in water, followed by evaporation to concentrate the salt solution, and finally, recrystallization to obtain sodium chloride crystals. Distillation is a process of separating components based on differences in their boiling points and is not commonly used for isolating sodium chloride from seawater.

12. Correct option: (A)
Among the given options, vegetable oil provides the maximum energy. Vegetable oils are high in fats, which are a concentrated source of energy. Fats yield more energy per gram

compared to carbohydrates (like glucose), proteins (found in cereals and pulses), and other nutrients.

13. Correct option: (A)

Cement mortar is commonly used for plastering. Plastering is the process of covering walls and ceilings with a layer of plaster, and cement mortar, which is a mixture of cement, sand, and water, is often employed for this purpose. While concrete and reinforced cement concrete (RCC) are used for structural purposes, they are not typically used directly for plastering.

14. Correct option: (D)

All the statements are correct:

Thermoplastics can be processed repeatedly because they soften when heated and harden when cooled without undergoing a chemical change.

Thermosetting plastics, on the other hand, cannot be softened by heating after they have undergone the curing or setting process, as they undergo a chemical change during this process.

Thermoplastics do not undergo a chemical change on heating; they retain their original chemical structure.

15. Correct option: (B)

Starch is a complex carbohydrate composed of carbon, hydrogen, and oxygen. The general chemical formula for starch is $(C_6H_{10}O_5)_n$, where "n" represents a large number, indicating the repeating units of the carbohydrate structure. Starch is a polysaccharide and is a common source of energy in the diet.

16. Correct option: (D)

Sodium lauryl sulphate is a detergent. It is a surfactant commonly used in various cleaning and personal care products due to its ability to lower the surface tension of water and enhance the cleaning properties of the product.

sodium stearate, sodium chloride, and sodium carbonate are not typically used as detergents. Sodium stearate is a soap, sodium chloride is table salt, and sodium carbonate is washing soda.

17. Correct option: (C)

Mortar is a mixture used in masonry construction to bind building blocks or stones. It typically consists of cement, sand, and water. The cement binds the sand particles together when it hydrates, creating a strong and durable bond between the masonry units. The proportion of each component can vary depending on the specific application.

18. Correct option: (D)

Limestone is used in the manufacture of quicklime (calcium oxide), glass, and cement. The specific use of limestone depends on the process it undergoes:

Quicklime (Calcium oxide) is produced by heating limestone, and it is used in various applications, including as a flux in the production of iron and steel.

Limestone is a key raw material in the production of glass.

In the cement manufacturing process, limestone is one of the main raw materials used to produce clinker, which is then ground to make cement.

19. Correct option: (D)

Iodized salt typically contains sodium iodide or potassium iodide, not lithium iodide. The purpose of adding iodine compounds to salt is to prevent iodine deficiency disorders and promote better public health.

20. Correct option: (C)

Potassium hydroxide (KOH) is used as one of the raw materials for making soft soap. The process involves saponification, where potassium hydroxide reacts with fats or oils to

produce soap and glycerol. Soft soap is a soap product that is typically made with potassium hydroxide instead of sodium hydroxide, which is used in the production of hard soaps.

21. Correct option: (D)

The metallic compound involved in cloud seeding is silver iodide, which is often dispersed into clouds to encourage the formation of ice crystals and precipitation. While silver iodide is used, it's important to note that aluminium-based compounds, such as aluminium oxide, have also been proposed for cloud seeding purposes.

22. Correct option: (A)

The correct sequence for the manufacture of glass is:
Mixing of raw materials in a proper proportion (2)
Addition of cullet (1)
Addition of colouring pigments (3)
Annealing (4)

23. Correct option: (D)

The correct sequence for the manufacture of common salt is:
Evaporation of seawater (2)
Dissolution in water (4)
Addition of sodium iodide and potassium iodide (3)
Recrystallization (1)

24. Correct option: (D)

Metal that cannot be used in printed circuit boards is Zinc.

25. Correct option: (C)

Graphite is used as a pencil lead because it is soft.

26. Correct option: (D)

Argon gas is heavier than air, so it cannot be used to fill weather observation balloons, as it does not allow the balloons to reach greater heights.

27. Correct option: (D)

The correct sequence for the manufacture of sodium chloride from seawater is as follows:
(4) Collection of seawater
(2) Evaporation
(1) Making of a hot saturated solution
(3) Recrystallisation

28. Correct option: (D)

Radon is a radioactive noble gas, and its radioactivity makes it unsuitable for such applications. Helium, neon, and argon are non-radioactive noble gases and can be used for filling up electric bulbs.

29. Correct option: (B)

Silicon is a semiconductor and has poor electrical conductivity. This property is not usually relevant in the context of non-stick cookware.

30. Correct option: (D)

Thermoplastics can be processed repeatedly, and they do not undergo a chemical change on heating. On the other hand, thermosetting plastics cannot be softened by heating because they undergo a chemical change during the curing process, which makes them hard and infusible. So, all the given options are differences between thermoplastic and thermosetting plastics.

6. Air and Oxygen

Solutions

LEVEL I

1. Correct option: (B) Decreases

As altitude increases, there are fewer air molecules above, resulting in reduced pressure exerted on the surface.

2. Correct option: (B) troposphere

In the troposphere, the temperature generally decreases with altitude. The reason is the troposphere's gases absorb very little of the incoming solar radiation. Instead, the ground absorbs this radiation and then heats the tropospheric air by conduction and convection.

3. Correct option: (C) sulphur dioxide

Suphur dioxide (SO_2) is not a greenhouse gas, thus it does not contribute to global warming.

4. Correct option: (C) O2

Oxygen is the most abundant element in the earth crust. Silicon is the second most abundant element and aluminium is the 3rd.

5. Correct option: (A) alloying

Alloying is the process of combining a metal with one or more other elements, either metals or non-metals, to create a homogeneous mixture with improved properties.

6. Correct option: (C) H202

H2O2 is used for the laboratory preparation of oxygen as it requires no heat and the rate of evolution of oxygen is moderate and under control.

7. Correct option: (B) liquid oxygen

Liquid oxygen is molecular oxygen in liquid form. It was employed as an oxidizer in the first liquid-fueled rocket built by Robert H. Goddard in 1926, and it is being used today. So, is used as a supporter of combustion in rockets and space ships.

8. Correct option: (B) magnesium

Magnesium burn in an atmosphere of CO2. When magnesium burns in carbon dioxide, it forms magnesium oxide and carbon as a product.

9. Correct option: (B) CO2

Soda water or carbonated water is water that has been injected under pressure with carbon dioxide gas. Carbonated water has the chemical formula H_2CO_3. It is popularly used as a beverage.

10. Correct option: (B) dry ice

Solid carbon dioxide is known as dry ice.

It is used for the refrigeration of sea food due to the property that it melts slowly and it directly changes from solid to a gaseous state which prevents liquid formation in the surrounding.

11. Correct option: (A) deep-sea diving

In the sea, pressure is more, so more amount of oxygen gets dissolved in the blood. This excess oxygen when dissolved in the blood causes several health problems like nausea, dizziness, etc.

12. Correct option: (B) ozone

Ozone in the troposphere is considered a pollutant, contributing to smog. However, in the upper layers (specifically the stratosphere), it protects life on Earth by absorbing harmful ultraviolet radiation.

13. Correct option: (B) downward displacement of water and upward displacement of air, respectively

Oxygen is prepared in the laboratory by the decomposition of Hydrogen peroxide. The oxygen evolved is collected by downward displacement of water. Carbon dioxide is prepared in the laboratory. When dilute Hydrochloric acid (HCl) reacts with Calcium carbonate or Marble (CaCO3), Calcium chloride (CaCl2), water (H2O) are formed and Carbon dioxide is evolved. The Carbon dioxide evolved is collected by upward displacement of air.

14. Correct option: (D) formation of auroras

Formation of auroras involves collision of electrons with molecules of N2 ana O2 and emission of beautiful radiations. It does not result in formation of nitrates.

15. Correct option: (D) 45312

16. Correct option: (C) O2

Nitrogen is the major component of air and occupies only 78% of the total volume of air. It is chemically less reactive in nature and it dilutes the activity of Oxygen gas and prevents rapid burning.

17. Correct option: (D) nitrogen dioxide

The gases like Carbon dioxide, water vapors, Methane, Ozone, CFCs (Chlorofluorocarbon), etc., absorb the IR radiations. Thus, Nitrogen dioxide is not a greenhouse gas. It does not contribute to the greenhouse effect.

18. Correct option: (A) Na

When sodium metal or its salt is heated in Bunsen flame, its valence shell electron is excited to higher energy levels of absorption of energy. When the excited electron returns to the ground state, it emits the extra energy in the yellow region of the electromagnetic spectrum.

19. Correct option: (B) CO2

Global warming results in a gradual increase in the temperature of the earth's surface. The main cause of global warming is the increase in air pollutants. Carbon dioxide is the main contributor to global warming.

20. Correct option: (C) iron

Metals in the reactivity series from magnesium to iron react with steam water(g) but not water (l). The reaction forms the metal oxide and hydrogen gas. The reaction between iron and steam is reversible.

21. Correct option: (A) neon

It is highly unreactive in nature so it is called inert gas, therefore used in the advertisement of glow tubes as it emits a reddish-orange glow.

22. Correct option: (B) Pascal

It is defined as a force of one Newton per square meter N/m2. 1 atm = 101325 Pa

23. Correct option: (D) oxygen

Oxygen was previously referred to as "acid-forming" due to its ability to form acids when it combines with other elements.

24. Correct option: (B) stratosphere

Ozone present in the stratosphere protects life on Earth's surface by absorbing and filtering out harmful ultraviolet (UV) radiation from the Sun.

25. Correct option: (D) chromium

Chromium is commonly used in alloying iron to form stainless steel, which helps prevent corrosion.

26. Correct option: (A) low atmospheric pressure

At higher altitudes the atmospheric pressure is low. When a certain altitude is reached the pressure outside the pen is lesser than the air pressure inside the fountain pen. This difference in pressure makes the ink flow from high pressure to low pressure. Thus because of the reduction in the atmospheric pressure at higher altitudes the fountain pen leaks as the pressure of air is less outside than inside the pen.

27. Correct option: (B) air conditioners

Pollutant such as CFCs (Chlorofluorocarbon) that is released from refrigerators and air conditioners when reaches the stratosphere layer react with Ozone and depletes the ozone layer.

28. Correct option: (D) death of aquatic animals

Acid rain can lead to the acidification of water bodies, affecting aquatic life and causing harm or death to aquatic animals.

29. Correct option: (D) W

Since sodium, magnesium and potassium are active metals, they catch fire when they react with oxygen and continue to burn in an atmosphere of N_2 , while tungsten (W) which burns in presence of oxygen, does not continue to burn in an inert atmosphere of N_2 .

30. Correct option: (C) It is heavier than air and is highly soluble in water.

Carbon dioxide gas is collected by the upward displacement of air. Carbon dioxide gas is heavier than water and is highly soluble in water. The solubility of Carbon dioxide increases with an increase in pressure.

31. Correct option: (C) 21534

Inversion layer:

- The layer above the troposphere is called the stratosphere and it extends up to 50 km above the earth's surface. Above 25 km, temperature increases due to ultraviolet (UV) radiation absorption. This region of increase in temperature is called the inversion layer.

Convection currents:

- The layer of the atmosphere which extends up to 20 km above the earth's surface is called the troposphere. Convection currents are present in this layer.

Auroras:

- The layer of the atmosphere which extends up to 690-950 km above the earth's surface is called the thermosphere or ionosphere.
- Auroras appear in this layer. Auroras are the luminous streams that are visible from the earth's surface at northern and southern latitudes.

Harmful ionizing radiations:

- The layer of the atmosphere which extends up to 10,000 km above the earth's surface is called the exosphere. This layer absorbs harmful ionizing radiation.

Burning up of meteorites:

- The layer of the atmosphere which extends up to 85 km above the earth's surface is called the mesosphere. Meteorites burn up in this layer.

32. Correct option: (B) 3241

Step 1: In the first step, nitrogen combines with oxygen to form Nitric oxide
Step 2: Nitric oxide further reacts with oxygen to form Nitrogen dioxide
Step 3: Now, Nitrogen dioxide combines with water and oxygen to form Nitric acid.
Step 4: In the last step, Nitric acid reacts with metal carbonates to form corresponding metal nitrate, Carbon dioxide and water.

LEVEL – II

1. Correct option: (D) CO2 water, and energy

It is an exothermic reaction in which complex food (say carbohydrate or glucose) in the presence of oxygen gives water and carbon dioxide and a large amount of energy.

2. Correct option: (C) It is only the human beings who need energy.

All the statements (A, B, and D) are correct, and humans are not the only ones needing energy; various organisms require energy for survival.

3. Correct option: (A) Carbon dioxide

Carbon Dioxide released in the atmosphere creates a green house effect by trapping the heat radiated from the earth.

4. Correct option: (B) global warming

Excessive levels of carbon dioxide (CO2) in the atmosphere contribute to global warming by trapping more heat.

5. Correct option: (D) All of these

Rainfall, sunlight, and oxygen are crucial for the survival of plants and animals.

6. Correct option: (C) Nitrogen

Nitrogen makes up the highest proportion of gases in Earth's atmosphere, approximately around 78%.

7. Correct option: (B) Carbon dioxide

Green plants utilize carbon dioxide (CO2) along with water and sunlight during photosynthesis to produce their food.

8. Correct option: (C) Oxygen

Oxygen is the gas essential for human respiration, as it is used in the process of breathing and cellular respiration.

9. Correct option: (A) Lighter?

Heating air causes it to expand and become less dense, making it lighter and rise.

10. Correct option: (D) 13 km

On average, the height of the troposphere extends up to approximately 13 kilometers above Earth's surface.

11. Correct option: (B) decreases

The pressure decreases as altitude increases within the troposphere

12. Correct option: (A) Ozone

Ozone present in the stratosphere acts as a protective layer that absorbs and filters harmful ultraviolet (UV) rays from the Sun, thus protecting life on Earth.

13. Correct option: (D) All of These

14. Correct option: (A) Temperature

15. Correct option: (D) 100°C

At sea level, water boils at 100 degrees Celsius under standard atmospheric pressure.

16. Correct option: (B) Sea Level

Air pressure is highest at or near sea level due to the weight of the air above exerting pressure on the surface.

17. Correct option: (A) Loo

A "loo" is a hot, dry, and strong wind that blows during summer in the Indian subcontinent, particularly in the region of the Indo-Gangetic Plain.

18. Correct option: (A) Sea breeze

A sea breeze is a local wind that occurs during the day when cooler air from the sea moves towards the warmer land.

19. Correct option: (C) Precipitation

Precipitation encompasses various forms of water falling from the atmosphere, such as rain, snow, sleet, or hail.

Fill in the blanks with appropriate words:

20. Correct option: (B) atmosphere

The layer of gases enveloping the Earth is termed the atmosphere. It consists of various gases, including nitrogen, oxygen, carbon dioxide, and others.

21. Correct option: (C) oxygen

During photosynthesis, green plants convert carbon dioxide and water into glucose and oxygen. Oxygen is released as a byproduct into the atmosphere.

22. Correct option: (A) Carbon dioxide

An elevated concentration of carbon dioxide in the atmosphere, largely due to human activities such as burning fossil fuels, is influencing global weather patterns and contributing to climate change.

23. Correct option: (A) Thermosphere

The ionosphere is a region within the thermosphere where ions and electrons are abundant, playing a role in radio wave propagation and auroras.

24. Correct option: (B) rain gauge

A rain gauge is a meteorological instrument used to measure the amount of rainfall at a specific location.

25. Correct option: (D) temperature

Insolation (incoming solar radiation) significantly influences the distribution and variation of temperatures on Earth's surface.

State whether the given statements are true or false.

26. Correct option: (B) False

Hot air is less dense and lighter than cold air. When heated, air expands and becomes less dense.

27. Correct option: (A) true

The greenhouse effect maintains Earth's temperature at a habitable level by trapping heat in the atmosphere.

28. Correct option: (B) False

The ozone layer is primarily located in the stratosphere, not the ionosphere.

29. Correct option: (A) true

Lighter gases like helium and hydrogen can escape Earth's gravitational pull from the exosphere and move into space.

30. Correct option: (A) true

A weather vane, wind vane, or weathercock is an instrument used for showing the direction of the wind.

31. Correct option: (A) true

Air moves from high-pressure areas to low-pressure areas. The bigger the difference between the pressures, the faster the air will move from high to low pressure.

32. Correct option: (B) d, c, e, b, a

1. Temperature - (d) Thermometer
2. Pressure - (c) Barometer
3. Seasonal wind - (e) Change directions in different seasons
4. Radio waves transmission - (b) Thermosphere
5. Nitrogen from air - (a) Bacteria in soil

7. Fiber to Fabric

Solutions

LEVEL – I

1. Correct option: (A)

Natural fibers are threads obtained from plants and animals. Examples are cotton, coconut, husk fibers etc.

2. Correct option: (C)

The fibers obtained from plants and animals are thread like strands which are used to produce fabrics in general used in paper producing factories, cloths producing factories.

3. Correct option: (D)

Natural fibers are used to make a variety of products, including boxes (using materials like straw or bamboo), chairs (using materials like rattan or wicker) in home decoration, and clothes preparation (using materials like cotton, linen, wool, silk, etc.).

4. Correct option: (D)

Fibers are derived from plants and animals which are long thread like structures used to prepare various products. All of the options are examples of plant fibers. These natural plant fibers are commonly used in various industries, such as textile, paper, and packaging, due to their strength, durability, and other desirable properties.

5. Correct option: (C)

Among the given options plastic is not a plant fiber. It is a synthetic material produced from petrochemicals and is not obtained from any plant source. The other options (Flax, Jute, and Cotton) are all examples of plant fibers.

6. Correct option: (C)

Silk and Wool are animal fibers among the given options. Silk is produced by silkworms while Wool is obtained from the fleece of sheep and some other animals like goats (Cashmere and Mohair). These animal fibers are widely used in the textile industry for making various types of winter clothing. Remaining options are plant fibers.

7. Correct option: (B)

Buffalo does not give wool. Wool is mainly obtained from sheep, whereas other animals like goats (Cashmere and Mohair), yaks also provide certain types of wool. However, buffalo does not produce wool.

8. Correct option: (B)

Silk comes from silkworms. Silkworms are the larvae or caterpillars of the silk moth (Bombyx mori). They produce silk by spinning a cocoon made of silk fibers to protect themselves during the pupal stage of their life cycle. The silk fibers produced by the silkworms are used to make silk fabric, which is highly prized for its smooth texture and luxurious feel.

9. Correct option: (A)

Woolen fabrics have a greater bulk than cotton fabrics. Wool fibers are crimped and have a more three-dimensional structure compared to the flat and smooth structure of cotton fibers. This crimped structure gives wool bulkiness, which helps in trapping air and providing insulation in colder temperatures. Cotton fibers, being smoother and flatter, have less bulk and are more lightweight compared to wool.

10. Correct option: (A)

Woolen fabrics are known for their excellent insulation properties, making them ideal for keeping us warm during cold days. The crimped structure of wool fibers creates air pockets that trap heat, providing better insulation and retaining body warmth. During cold weather, wearing woolen clothing can help keep us cozy and comfortable.

11. Correct option: (D)
Wool comes from sheep. Sheep are domesticated animals that are raised for their wool, which is the fibrous coat that grows on their bodies. The process of obtaining wool from sheep involves shearing, where the wool is carefully cut off the sheep's body to produce various woolen products.
12. Correct option: (C)
Wool is obtained from the fleece (hair) of sheep, which is the natural covering of the sheep's body. The hair consists of wool fibers, and it is sheared off the sheep's body to yield the wool.
13. Correct option: (D)
Sheep: Sheep are the most common source of wool and produce a variety of wool types, depending on the sheep breed.
Goat: Goats provide several types of wool, including Mohair (from Angora goats) and Cashmere (Kashmir goats).
Yak: Yaks, which live in the Himalayan region, also produce a type of wool known as yak wool.
So, all three animals give wool.
14. Correct option: (B)
Air is a poor conductor of heat. Meaning it does not conduct heat well compared to materials like metals. Air has low thermal conductivity, which is why it is used as an insulating material in various applications. When there is a temperature difference between two objects, air acts as a barrier, reducing the transfer of heat between them.
15. Correct option: (D)
The Lohi breed of sheep is found in both Rajasthan and Punjab states in India. These sheep are adapted to the arid and semi-arid regions of these states and are known for their fine quality wool, which is used in the production of textiles and clothing.
16. Correct option: (D)
The Rampur Bushair breed of sheep are found in both Uttar Pradesh and Himachal Pradesh states of India. These states are known for their diverse sheep breeds, and Rampur Bushair is one of the popular sheep breeds found in these regions.
17. Correct option: (D)
The Nali breed of sheep is found in multiple states, including Rajasthan, Haryana, and Punjab. It is a popular breed known for its good wool quality and adaptability to the arid and semi-arid regions of these states. The Nali sheep are well-suited to the local climatic conditions and are reared for their wool.
18. Correct option: (A)
The Marwari breed of sheep is found in the state of Gujarat, India. It is a breed that is well-adapted to the arid and semi-arid regions of Gujarat and is known for its ability to thrive in such harsh environments. Marwari sheep are primarily reared for their fine quality of wool, which is of good quality and suitable for various textile applications.
19. Correct option: (A)
The Patanwadi breed of sheep is found in the state of Gujarat (coastal plains of the Saurashtra and Kutch) in India. It is a popular Patanwadi sheep are primarily raised for their wool and are well-suited to the semi-arid and arid environments of Gujarat.
20. Correct option: (D)
The Rampur Bushair breed of sheep has brown fleece wool. This breed is found in Uttar Pradesh and Himachal Pradesh states of India. The wool from Rampur Bushair sheep can vary in color and is known for its good quality and unique brown hue.

21. Correct option: (C)

Wool obtained from sheep and most other wool-producing animals. The natural color of wool is generally creamy white. However, it's important to note that wool can come in various natural colors, including shades of brown, gray, and black, depending on the breed of the sheep or the animal it comes from.

22. Correct option: (C)

Yak wool is commonly found in the hilly regions of both Tibet and Ladakh. Yaks are primarily found in the high-altitude regions of the Himalayas, including parts of Tibet and the Ladakh region of India. The wool obtained from yaks, known as yak wool, is warm, durable, and suitable for making clothing and textiles that provide insulation in cold and harsh climates.

23)	**A**	**24)**	**D**	**25)**	**B**	**26)**	**D**	**27)**	**C**	**28)**	**B**	**29)**	**B**
30)	**C**	**31)**	**C**	**32)**	**C**	**33)**	**C**	**34)**	**A**	**35)**	**D**	**36)**	**B**
37)	**D**	**38)**	**A**	**39)**	**A**	**40)**	**A**						

LEVEL – II

1. Correct option: (A)

Rearing of sheep means to look after the sheep by providing them with food, shelter and healthcare. It involves the management and care of the sheep to ensure their well-being and productivity.

2. Correct option: (B)

The persons who look after the sheep are called shepherds. They look after herding and caring for the sheep, ensuring that they have access to food, water, and shelter, as well as protecting them from predators and guiding them to suitable grazing areas. Shepherds play a crucial role in sheep farming and ensuring the well-being of the flock.

3. Correct option: (A)

Sheep are herbivores, meaning they primarily feed on plant material. They are animals, equipped with a four-chambered stomach that allows them to digest fibrous plant material efficiently. Sheep graze on grass, leaves, and other vegetation.

4. Correct option: (A)

In winter, sheep are often kept indoors or in closed sheltered areas to protect them from the cold weather. Sheep are susceptible to adverse weather conditions, specially extreme cold and wetness, which can cause health issues and even death in severe cases. By keeping them indoors or providing them with shelter, farmers can ensure that the sheep are safe, warm, and protected from the harsh winter elements.

5. Correct option: (A)

Cutting of wool coat of sheep along with thin layer of skin is called as **Fleece**.

6. Correct option: (A)

Shearing is the process of removing the wool from the body of a sheep in the form of a fleece. This is essential for the health and well-being of the sheep as it prevents them from becoming too hot during warm weather and reduces the risk of certain skin conditions. The shearing process is typically performed by skilled workers called shearers or wool harvesters. Shearing allows farmers to collect wool, which is a valuable natural fiber used to produce cloths in various textiles and products.

7. Correct option: (A)

The process of washing the fleece (cut hair of sheep) to remove dust, dirt, dried sweat, and grease is called scouring. Scouring is an essential step in processing of raw wool, as it prepares the wool for further manufacturing into various products such as yarn, fabric, and garments.

These impurities include lanolin which is a natural oil secreted by the sheep's skin, dust and dirt from the environment, and sweat residues. If not properly removed, these impurities can affect the quality and appearance of the final wool product.

8. Correct option: (C)

The fleece of sheep is washed with water and soap during the scouring process. Soap helps to remove dirt, dust, dried sweat, and grease from the wool fibers, leaving the fleece clean and ready for further processing. The use of soap in the scouring process ensures that impurities are effectively removed, and the wool is prepared for manufacturing into various woolen products like yarn, fabric, and garments.

9. Correct option: (A)

The process of separating the fleece of a sheep into sections according to the quality of woolen fibers is called sorting. After shearing and before the scouring process, the raw wool is sorted to classify the fibers based on characteristics such as fiber length, fineness, color, and overall quality.

Sorting is an essential step in the wool processing chain, as it allows the wool to be categorized into different grades, each suitable for specific products.

10. Correct option: (D)

Silk fiber is made of protein. It is primarily composed of a protein called fibroin, which makes up the important part of the silk fiber. The fibroin protein provides silk with its unique properties such as strength, smoothness, and luster.

Silk is produced by various insects, but the most well-known and widely used silk is produced by the silkworm (Bombyx mori). The silkworm spins a cocoon made of silk fibers to protect itself during the pupal stage of its life cycle. The silk fibers are produced from specialized glands in the silkworm's head called spinnerets.

11. Correct option: (B)

Silk is one of the strongest natural fibers. It has an impressive tensile strength, means it can withstand considerable pulling or stretching forces without breaking. Despite its delicate and smooth appearance, silk is surprisingly durable and has been historically used in various applications, including garments, textiles, and even parachutes in some cases.

12. Correct option: (C)

Sericulture is an ancient practice that involves the cultivation of silkworms and the harvesting of their cocoons to produce silk fibers. It is an essential part of the silk production process.

Sericulture involves several stages, starting with the production of silkworm eggs, which are carefully incubated to hatch into larvae. The larvae are then fed with mulberry leaves. They undergo several molting stages, during which they shed their skin and grow larger.

After reaching the final larval stage, the silkworms spin a cocoon made of silk fibers to protect themselves during the pupal stage of their life cycle. The cocoons are collected and then processed to obtain the raw silk fibers.

13. Correct option: (A)

The silky covering spun by the larva of silkworm, is called a cocoon. When the silkworm is ready to undergo the pupal stage of its life cycle, it spins a protective casing around itself using silk fibers produced from specialized glands in its head. This cocoon serves to protect the silkworm during its transformation into a pupa, and later, into an adult moth.

14. Correct option: (D)

The cocoon is made by the silkworm to protect its development as a pupa. After going through several larval stages, the silkworm spins a silk cocoon around itself during its pupal stage. Inside the cocoon, the silkworm undergoes metamorphosis and transforms into an adult moth. The cocoon serves as a protective casing that shields the pupa from external threats until it is ready to emerge as an adult moth.

15. Correct option: (C)
The breaking of the cocoon lowers the quality of silk. When the cocoon is intact and not broken, the silk fibers inside it can be carefully unraveled in long continuous filaments. This results in high-quality silk, which is valued for its strength, sheen, and smooth texture.
However, if the cocoon is broken, the long silk filaments are cut into shorter segments, which reduces the overall quality of the silk. Silk with shorter fibers may be less durable and have a rougher texture, making it less desirable for certain high-end silk products.

16. Correct option: (B)
The process of taking out silk fibers from the cocoons for use as silk is called reeling. Reeling is an essential step in the silk production process and involves carefully unwinding the silk filaments from the cocoons without breaking them.
During reeling, the cocoon is softened by exposing it to heat or steam. Then, skilled workers gently and carefully unwind the silk filaments from the cocoon, sometimes using a special tool called a reel.

17. Correct option: (A)
The most common silk moth is Mulberry silk moth, scientifically known as Bombyx mori. Mulberry silk is widely produced and is obtained from the domesticated silkworms of the species **Bombyx mori**.
These silkworms feed primarily on the leaves of the mulberry tree, which is why the silk produced by them is called Mulberry silk. The cultivation of silkworms for Mulberry silk production is a practice known as sericulture.

18. Correct option: (D)
Rayon is a man-made fiber that is often used as a substitute for natural silk in textiles and clothing. It is produced from cellulose, usually obtained from wood pulp, and then chemically processed to create long filaments that are spun into threads.

19. Correct option: (B)
The burning test is one of the methods used to distinguish difference between natural silk and artificial silk. This test involves burning a small sample of the fabric to observe its behavior and characteristics during combustion.
Natural Silk (Pure Silk): When natural silk burns, it will have a smell similar to burning hair or feathers. The flame will be steady. Once the flame is removed, the silk will stop burning on its own. The ash left behind will be very fine and powdery, often white or light gray in color.
Artificial Silk (Rayon): When rayon burns, it may produce a chemical or sweet smell. The flame will burn steadily and is easier to maintain compared to natural silk. After the flame is removed, rayon will continue to burn for a short time. The ash left behind is typically darker and may be bead-like.

20. Correct option: (C)
China is the leading manufacturer of silk in the world. The country has a long history of silk production, dating back thousands of years. Chinese silk is highly regarded for its quality. China's favorable climate, increase supply of mulberry leaves (the primary food of silkworms), and expertise in silk production have contributed to its dominant position in the global silk industry.

21)	**B**	**22)**	**A**	**23)**	**C**	**24)**	**D**	**25)**	**B**	**26)**	**B**	**27)**	**A**
28)	**A**	**29)**	**B**	**30)**	**B**	**31)**	**C**	**32)**	**A**	**33)**	**A**	**34)**	**A**
35)	**D**	**36)**	**C**	**37)**	**B**	**38)**	**D**	**39)**	**ABC**	**40)**	**AB**	**41)**	**AB**
42)	**ABCD**	**43)**	**AB**	**44)**	**ABC**	**45)**	**ABCD**	**46)**	**ABCD**				

47) a – S, b – P, c – Q, d – R